SOFTWARE REQUIREMENTS SPECIFICATION

CampusConnect: University Club, Event & Activity Management Platform

Attribute	Specification Detail
Document Identifier	CC-SRS-2026-V2.0 (IEEE Std 830-1998 / ISO/IEC/IEEE 29148 Standard)
Engineering Version	Version 2.0 (Formal Baseline)
Software Engineering Team	Md. Shakhawat Hossain (Student ID: 0432320005101009) | Fahim Hossain (Student ID: 0432320005101007) | Department of CSE
Course Advisor & Supervisor	Al Imtiaz | Department of Computer Science and Engineering
Reviewing Body	Faculty Academic Review Board | Division of Computing and Systems
Release Date & Status	October 2026 | Approved for Engineering Implementation

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) establishes the complete functional, non-functional, data, and system interface requirements for CampusConnect. This document defines the engineering baseline and functions as an authoritative contract among university stakeholders, student organizations, faculty advisors (Al Imtiaz), and the software development team led by Md. Shakhawat Hossain (Student ID: 0432320005101009) and Fahim Hossain (Student ID: 0432320005101007).

1.2 Document Conventions

Requirements are structured using consistent formal identifiers: FR-XX designates functional requirements, NFR-XX designates non-functional quality attributes, and AC-XX designates verifiable acceptance criteria. Requirement priority is categorized as High (mandatory core functionality), Medium (secondary operational feature), or Low (enhancement). The document adheres strictly to standard ASCII hyphens (-), colons (:), and pipes (|) without typographic long hyphens.

1.3 Intended Audience and Reading Suggestions

This specification is organized for distinct technical and managerial stakeholders:

- **Software Developers & System Architects:** Focus on Section 3 (System Features), Section 4 (Interfaces), Section 5 (NFRs), and Section 6 (Relational Schema).

- **Faculty Advisors & University Administrators:** Focus on Section 1.4 (Scope), Section 2.3 (User Classes), Section 3.3 (Event Governance), and Section 7 (RBAC Matrix).
- **Quality Assurance Engineers & Testers:** Focus on Section 3 (Normal and Exception Flows), Section 5 (Performance Benchmarks), and Section 8 (Traceability & Acceptance Criteria).

1.4 Project Scope

CampusConnect replaces fragmented communication channels (informal messaging threads, social media groups, and physical flyers) and error-prone spreadsheets with an integrated, auditable web application platform.

- **In-Scope Capabilities:** User profile management, secure authentication, club directories, membership requests, multi-tier event proposals, faculty advisor review workflows, concurrency-safe registrations, waitlist management, dynamic rotating QR attendance tracking, digitally signed PDF certificate issuance with public verification, in-app notices, and institutional analytics.
- **Out-of-Scope (Release 1.0):** Online ticket payment gateway integration, native mobile operating system builds (iOS/Android native binaries), biometric hardware scanner integration, and automated synchronization with legacy Student Information Systems (SIS).

1.5 Definitions, Acronyms, and Abbreviations

Acronym / Term	Definition
ACID	Atomicity, Consistency, Isolation, Durability guarantees in transactional database engines.
Argon2id	Key derivation function optimized to resist GPU cracking and side-channel memory attacks.
JWT	JSON Web Token (RFC 7519) utilized for compact, stateless claims verification.
RBAC	Role-Based Access Control restricting system operations based on assigned organizational roles.
TOTP	Time-Based One-Time Password algorithm (RFC 6238) utilized for rotating attendance tokens.
UUIDv4	Universally Unique Identifier version 4 generated via cryptographically secure pseudo-random numbers.

1.6 Project Engineering Team and Governance

The design, specification, and implementation of CampusConnect is executed by the project engineering team with academic supervision:

Project Role	Name & Identification	Department & Affiliation	Engineering Scope & Responsibilities
Team Leader & Engineering Lead	Md. Shakhawat Hossain (ID: 0432320005101009)	Department of Computer Science and Engineering (UITS)	System architecture, relational data modeling, secure authentication, dynamic QR attendance engine, and tamper-proof PDF certificate issuance.
Software Engineer & Team Member	Fahim Hossain (ID: 0432320005101007)	Department of Computer Science and Engineering (UITS)	Core feature implementation, test engineering, dynamic QR verification integration, and technical documentation.
Course Advisor & Supervisor	Al Imtiaz	Department of Computer Science and Engineering (UITS)	Requirements governance, technical specification review, academic standards compliance, and engineering baseline sign-off.

1.7 References

1. IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications.

2. ISO/IEC/IEEE 29148:2018: Systems and software engineering: Life cycle processes: Requirements engineering.

3. RFC 2119: Key words for use in RFCs to Indicate Requirement Levels.

4. RFC 7519: JSON Web Token (JWT) Specification.

5. W3C Web Content Accessibility Guidelines (WCAG) 2.1 Level AA.

2. Overall Description

2.1 Product Perspective

CampusConnect operates as a standalone multi-tenant web application hosted within the university cloud environment. It coordinates activities across academic departments, student clubs, and administrative offices through a decoupled client-server architecture. The presentation layer communicates with the backend application server exclusively via a secure RESTful API.

2.2 Product Functions Summary

The system delivers seven principal business capabilities: (1) Identity and Credential Governance; (2) Club Directory and Membership Lifecycle; (3) Event Lifecycle and Multi-Tier Approval; (4) Concurrency-Safe Event Registration and Waitlisting; (5) Anti-Proxy Dynamic

QR Attendance Tracking; (6) Cryptographically Signed PDF Certificate Issuance; and (7) Executive and Institutional Analytics Reporting.

2.3 User Classes and Characteristics

User Role	Target User Persona	Primary System Responsibilities
Student	Enrolled university undergraduate and graduate students.	Explore clubs and events, register, present live QR passes, and download certificates.
Club Executive	Elected student leaders managing registered campus organizations.	Draft events, manage member applications, track participant rosters, and record attendance.
Faculty Advisor	Appointed academic faculty members supervising student clubs.	Audit club compliance, review draft events, and approve or reject proposals with remarks.
Platform Admin	Central IT staff and Student Affairs officers.	Oversee platform categories, audit security logs, override roles, and generate campus reports.

2.4 Operating Environment

Server Infrastructure: Linux (Ubuntu 22.04 LTS / 24.04 LTS), Node.js runtime (v20+ LTS), PostgreSQL 16 relational database, and Redis 7 in-memory cache. Client Infrastructure: Modern desktop and mobile browsers supporting ECMAScript 2022+ (Google Chrome 115+, Mozilla Firefox 115+, Apple Safari 16+, and Microsoft Edge 115+). Responsive layouts support viewports from 360px width upwards.

2.5 Design and Implementation Constraints

- **Relational Engine Mandate:** PostgreSQL 16 MUST be the persistent database engine to guarantee ACID transactions and row-level locking (SELECT ... FOR UPDATE) during high-concurrency event registration bursts.
- **Type Safety:** TypeScript MUST be used across frontend and backend layers to share data models and eliminate runtime type mismatches.
- **Typography Standard:** All application logs, UI text, notification templates, and generated documents MUST strictly avoid em-dashes and en-dashes.

2.6 Assumptions and Dependencies

Users possess devices with web browser access and HTTPS connectivity. Executives conducting QR check-ins require camera-enabled devices supporting the HTML5 MediaDevices API. Automated email notifications depend on institutional SMTP availability.

3. System Features and Functional Requirements

3.1 User Authentication and Profile Management

FR-01: User Provisioning. Prospective users create accounts with institutional email (@university.edu), full name, student ID, department, and password (minimum 8 characters with mixed case, number, and symbol). Passwords are hashed using Argon2id. An email verification token (UUIDv4) is issued with a 24-hour expiration window.

FR-02: Authentication & Session Management. Users authenticate via email and password. Protected endpoints enforce rate limits (maximum 5 failed attempts per 15 minutes per IP). On successful verification, the backend issues a short-lived JWT access token (15-minute validity) and a refresh token (7-day validity) transmitted exclusively via HttpOnly, Secure, SameSite=Strict cookies.

FR-03: Token Revocation. Explicit logout revokes the active refresh token in the database and clears client cookies.

FR-04: Profile Management. Students update permitted profile fields (contact phone, bio, notification preferences). Primary identifiers (email, student ID, and role) remain immutable.

3.2 Club Directory and Membership Lifecycle

FR-05: Club Directory. Searchable directory categorizing active university organizations with details on mission, executive leadership, advisor, and active events.

FR-06: Membership Application. Enrolled students submit membership applications with a statement of interest. Duplicate pending applications are rejected by database unique constraints.

FR-07: Executive Processing. Club executives review pending applications and record decisions (Approved or Rejected). System logs the approving executive ID and timestamp.

FR-08: Executive Succession. Faculty advisors or administrators promote active club members to executive rank or revoke leadership upon term completion.

3.3 Event Creation, Lifecycle, and Approval Workflow

FR-09: Event Drafting. Executives draft events specifying title, description, category, venue, start/end timestamps, seat capacity, registration deadline, and prerequisites. State is set to DRAFT.

FR-10: Proposal Submission. Executive submits draft event for formal academic review. Status transitions from DRAFT to PENDING_APPROVAL. Event editing is locked during review.

FR-11: Faculty Advisor Review. Designated faculty advisors review proposal details. Approval transitions status to APPROVED and enables public publishing. Rejection transitions status to REJECTED, mandates written feedback remarks, and returns the event to an editable draft state.

FR-12: Event Cancellation. In emergencies, executives or advisors cancel scheduled events. System marks event as CANCELLED and dispatches immediate notification alerts to all registered attendees.

3.4 Event Discovery, Registration, and Concurrency Control

FR-13: Event Discovery Engine. Full-text search and multi-facet filtering by club, category, date, and open seat availability with p95 latency under 200 milliseconds.

FR-14: Concurrency-Safe Registration. Students register for published events before registration deadlines. The registration engine executes inside a PostgreSQL READ COMMITTED transaction with row-level locking (SELECT capacity, registered_count FROM events WHERE id = $1 FOR UPDATE). If registered_count < capacity, a CONFIRMED registration is created and registered_count incremented atomically. If capacity is exhausted, the user is offered waitlist placement.

FR-15: Automated Waitlist Management. When an event reaches maximum capacity, applicants join an ordered waitlist (WAITLISTED). Upon cancellation by a confirmed participant, the system automatically promotes the first waitlisted applicant to CONFIRMED and dispatches an urgent confirmation alert.

FR-16: Roster Export. Authorized club executives export registration rosters with participant names, student IDs, and registration timestamps in CSV format.

3.5 Secure Attendance Tracking and Dynamic QR Engine

FR-17: Dynamic Rotating QR Tokens. To prevent attendance fraud via screenshotted QR passes, organizer terminals project a dynamic QR code refreshed every 20 seconds using HMAC-SHA256(event_id + time_slot, secret_key). Students scan the live display using their camera.

FR-18: Attendance Validation. The server validates the token against the active and immediate preceding 20-second time slots to tolerate network latency. Upon validation, an attendance record is created with method QR_SCAN, and the registration record is updated to ATTENDED. Duplicate scans are rejected.

FR-19: Manual Attendance Fallback. If a student lacks a functional device or experiences camera failure, authorized executives manually record attendance using the student ID. The system records method MANUAL_OVERRIDE alongside the executive user ID in the audit log.

3.6 Digital Certificate Issuance and Public Verification

FR-20: Eligibility Verification. Attendees are eligible for certificates if they hold an attendance record of PRESENT, have submitted the required post-event feedback form, and the event has concluded.

FR-21: Tamper-Resistant PDF Generation. An asynchronous worker generates a PDF certificate featuring the recipient name, event title, completion date, unique serial number (CC-YYYY-

[HEX]), and a cryptographic SHA-256 fingerprint. An embedded verification QR code links directly to the public verification endpoint.

FR-22: Public Verification Portal. A publicly accessible endpoint (GET /api/v1/certificates/verify/:serial_number) allows academic boards, employers, and external evaluators to confirm certificate authenticity without requiring authentication credentials.

3.7 Notification Dispatch and Announcements

FR-23: Broadcast Announcements. Club executives publish notices to club members or registered event attendees via in-app feeds and broadcast emails.

FR-24: Notification Center. Delivers real-time in-app alerts with unread badge counters for registration updates, event approvals, waitlist promotions, and schedule changes.

3.8 Post-Event Feedback and Analytics

FR-25: Structured Feedback. Attendees submit numerical ratings (1 to 5 stars) and qualitative commentary post-event. Duplicate submissions per event are prevented.

FR-26: Analytics Dashboards. Executives view club-level attendance rates, registration velocity, and satisfaction scores. Administrators access university-wide metrics.

3.9 Governance and Immutable Audit Logging

FR-27: Immutable Audit Logging. Every administrative event (approval, rejection, role change, manual attendance override, or certificate revocation) is permanently recorded in the audit_logs table with actor ID, timestamp, IP address, and state snapshot.

4. External Interface Requirements

4.1 User Interfaces

The interface utilizes responsive layouts conforming to WCAG 2.1 Level AA accessibility standards, featuring high text contrast (minimum 4.5:1), keyboard-navigable controls, and descriptive semantic markup across desktop and mobile devices.

4.2 Hardware Interfaces

Client devices utilize camera hardware via standard HTML5 MediaDevices APIs to scan dynamic QR attendance tokens.

4.3 Software Interfaces

Component	Technology Specification	Interaction Purpose
Relational Database	PostgreSQL 16 via pooled TCP connection	ACID storage for users, clubs, events, registrations, and audit logs.
In-Memory Cache	Redis 7 via RESP protocol	API rate limiting, session token storage, and dynamic QR nonces.
Asynchronous Queue	BullMQ background worker system	Decoupled rendering of PDF certificates and bulk email dispatching.
Email Delivery	Configured institutional SMTP gateway	Transactional verification emails and broadcast announcements.

4.4 Communications Interfaces

All network traffic MUST use HTTPS over TLS 1.3. Plain HTTP connections are automatically redirected with HTTP Strict Transport Security (HSTS) headers enabled. RESTful endpoints exchange JSON payloads using standard HTTP status codes.

5. Non-Functional Requirements

ID	Category	Quantitative Requirement Specification	Verification Method
NFR-01	Performance	95% of read requests must respond within < 250ms under 500 concurrent users.	Automated k6 load test
NFR-02	Performance	95% of write transactions (registration, attendance) must complete in < 400ms.	Load test under peak load
NFR-03	Throughput	Certificate generation and email dispatch completed in < 60s asynchronously.	Background queue benchmark
NFR-04	Availability	Platform maintains >= 99.5% uptime during standard university academic terms.	Cloud uptime monitor
NFR-05	Reliability	Zero event over-registration permitted under concurrent race conditions.	Race condition stress test
NFR-06	Security	Passwords hashed via Argon2id (memory: 64MB, iterations: 3, parallelism: 1).	Security code audit
NFR-07	Security	Session tokens stored in HttpOnly, Secure, SameSite=Strict cookies.	Network inspector test
NFR-08	Security	Authentication routes enforce maximum 5 failed attempts per 15-minute window.	Brute force test suite
NFR-09	Compliance	All administrative operations recorded in immutable audit_logs table.	Database audit log review
NFR-10	Recovery	Daily pg_dump with WAL archiving supporting RPO < 1 hour and RTO < 4 hours.	Disaster recovery drill

6. Data Architecture and Relational Schema

CampusConnect mandates PostgreSQL 16 to guarantee relational integrity, foreign key cascading, and row-level locking. The schema comprises 12 normalized tables:

Table Name	Primary Key	Key Foreign Keys	Critical Constraints & Indexes
users	id (UUID)	None	UNIQUE(email), role ENUM, status ENUM
student_profiles	id (UUID)	user_id -> users(id)	UNIQUE(user_id), UNIQUE(student_id_number)
clubs	id (UUID)	faculty_advisor_id -> users(id)	UNIQUE(name), UNIQUE(slug)
club_memberships	id (UUID)	club_id -> clubs(id), user_id -> users(id)	UNIQUE(club_id, user_id), status ENUM
events	id (UUID)	club_id -> clubs(id), created_by -> users(id)	CHECK(capacity > 0), registered_count <= capacity
registrations	id (UUID)	event_id -> events(id), user_id -> users(id)	UNIQUE(event_id, user_id), UNIQUE(ticket_code)
attendance_records	id (UUID)	registration_id -> registrations(id)	UNIQUE(registration_id), method ENUM
certificates	id (UUID)	event_id -> events(id), user_id -> users(id)	UNIQUE(serial_number), UNIQUE(event_id, user_id)
announcements	id (UUID)	club_id -> clubs(id), author_id -> users(id)	Index on (club_id, created_at DESC)
notifications	id (UUID)	recipient_id -> users(id)	Index on (recipient_id, is_read)
feedback_submissions	id (UUID)	event_id -> events(id), user_id -> users(id)	UNIQUE(event_id, user_id), CHECK(rating BETWEEN 1 AND 5)
audit_logs	id (BIGSERIAL)	actor_user_id -> users(id)	Immutable append-only, index on (target_entity, target_id)

7. Role-Based Access Control (RBAC) Matrix

System Operation / Resource	Student	Club Executive	Faculty Advisor	Platform Admin
Browse Clubs and Public Events	Read	Read	Read	Read
Apply for Club Membership	Create	N/A	N/A	N/A
Manage Club Members	None	Approve / Reject	Read	Full Control
Draft and Edit Event Proposals	None	Create / Update (Own)	Read	Full Control
Review and Approve / Reject Events	None	None	Approve / Reject	Full Control
Register / Cancel Event Attendance	Create / Cancel	Create / Cancel	Read	Full Control
Conduct Dynamic QR Attendance Scan	None	Execute (Own Club)	Read	Full Control
Execute Manual Attendance Override	None	Execute (With Log)	Execute	Full Control
Download Earned Certificate PDF	Read (Own)	Read (Own)	Read	Full Control
Verify Certificate Validity (Public)	Read	Read	Read	Read
Broadcast Club Announcements	None	Create (Own Club)	Create (Assigned)	Create (Global)
Inspect System Audit Logs	None	None	None	Read
Assign User Roles and Permissions	None	None	None	Full Control

8. Traceability Matrix and Acceptance Criteria

8.1 Requirements Traceability Matrix

Business Objective	Functional Reqs	Relational Entities	Acceptance Criteria
Centralize Activity Discovery	FR-05, FR-13	clubs, events	AC-01, AC-02
Secure Identity Governance	FR-01, FR-02, FR-03	users, audit_logs	AC-03, AC-04
Event Approval & Moderation	FR-09, FR-10, FR-11	events, audit_logs	AC-05, AC-06
Concurrency-Safe Registration	FR-14, FR-15	registrations, events	AC-07, AC-08
Tamper-Proof Attendance Tracking	FR-17, FR-18, FR-19	attendance_records	AC-09, AC-10
Verifiable Digital Credentials	FR-20, FR-21, FR-22	certificates	AC-11, AC-12

8.2 Executable Acceptance Scenarios (BDD Format)

Scenario AC-07: Concurrency-Safe Capacity Enforcement

```
Scenario: Enforce maximum registration capacity under concurrent requests
  Given an event 'AI Innovation Workshop' exists with capacity 50 and
registered_count 49
  When two students attempt to register simultaneously at timestamp T
  Then exactly one student registration succeeds with HTTP 201 Created
  And the second student receives HTTP 200 OK with status 'WAITLISTED'
  And the event registered_count remains exactly 50
  And no database lock deadlock error is exposed to either client
```

Scenario AC-09: Defense Against Attendance QR Code Screenshot Sharing

```
Scenario: Reject expired or forwarded attendance QR token
  Given an ongoing event 'Robotics Hackathon'
  And student 'Rahim' holds a confirmed registration
  When student 'Karim' attempts check-in using a screenshot of Rahim's QR token
taken 45s prior
  Then the check-in request is rejected with HTTP 400 Bad Request
  And the error response specifies 'QR token expired. Please rescan live
display.'
  And no record is created in the attendance_records table
```

Scenario AC-11: Public Certificate Verification

```
Scenario: Public verification of earned completion certificate
  Given student 'Ayesha' has status 'ATTENDED' and completed event feedback
  When an external evaluator visits '/api/v1/certificates/verify/CC-2026-
```

```
A1B2C3D4'
  Then the system returns recipient name, event title, issuing club, and issue
date
  And the certificate cryptographic status is confirmed as 'VALID'
  And no login credentials are required to view the verification status
```

9. Verification and Quality Assurance Plan

1. Unit Testing: Core utilities (Argon2id password hasher, HMAC dynamic token generator, certificate serial generator) and schema validation middleware tested using Vitest / Jest targeting >= 85% statement coverage.

2. Integration Testing: End-to-end API workflows tested against an ephemeral PostgreSQL Docker test instance using Supertest.

3. Concurrency Load Testing: Simulated registration rushes executed via k6 (200 virtual users competing for the final 5 event slots) to certify zero over-booking and zero deadlocks.

4. Security Audit: Static code analysis via ESLint security plugins and dynamic OWASP Top 10 penetration sweeps.